

$$mv^2/r = mg \rightarrow v^2 = (23)(9.81) \rightarrow v = 15$$

8 Answer: B

$$\text{At A, GPE} = U_A = -\frac{GMm}{(h_1+R)}$$

$$\text{At B, GPE} = U_B = -\frac{GMm}{(h_2+R)}$$

$$\text{The increase in GPE} = U_B - U_A = GMm \left(\frac{1}{R+h_1} - \frac{1}{R+h_2} \right)$$